



BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) - 2024
SEMESTER –V

Course Code: BCA59114

Course Name: Design and Analysis of Algorithm Lab

Weekly Contact Hours: 4P

Credits: 2

Total Allotted Hours: 60P

Course Objective: This course aims to equip students to write programs with different algorithms and implement them to solve real world problems.

Pre-requisite(s): Basic programming knowledge.

Course Outcomes (COs): After completion of the course students will be able to:

CO1: Develop programs to implement recursive functions to solve different problems and analyze the efficiency.

CO2: Develop programs for sorting a given set of elements and analyze its time complexity.

CO3: Solve and analyze the problems using brute force algorithm, greedy strategy, dynamic programming and backtracking strategy.

CO4: Evaluate graph theory algorithms to solve various problems.

CO5: Develop programs on string matching algorithms and analyze them.

Module 1: Recursion

[4H]

Programs to familiar with recursion-tower of Hanoi, permutations and combinations, recursive maxmin,

Module 2: Searching and Sorting Algorithm

[8H]

Programs of Linear Search and Binary search – their comparisons, Interpolation search and its complexity. Merge sort, quick sort and heap sort, their complexity analysis in best case, worst case and average case.

Module 3: Algorithm Design Techniques

[24H]

Writing programs on Brute force algorithm; divide and conquer strategy - matrix multiplication using divide and conquer and Strassen's algorithm, greedy techniques - fractional knapsack problem, Job sequencing with deadline. Dynamic programming - 0/1 knapsack problem, matrix chain multiplication. Backtracking-N-queens Problem.

Module 4: Graph and Tree Algorithms

[20H]

Programs on Graph traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); shortest path algorithms Dijkstra's algorithm, Bellman Ford algorithm, Floyd-Warshall algorithm, spanning tree Prim's and Krushkal algorithm.



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Module 5: String Matching

[4H]

String matching algorithm – naïve algorithm, KMP algorithm.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K5, K6
2.	CO2	M2	K4, K6
3.	CO3	M3	K4, K5
4.	CO4	M4	K5, K6
5.	CO5	M5	K4, K6

Text books:

1. Introduction to Algorithms, Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022), MIT Press.

Reference books:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Sartaj Sahni and S. Rajasekharan, Universities Press, 2nd edition.